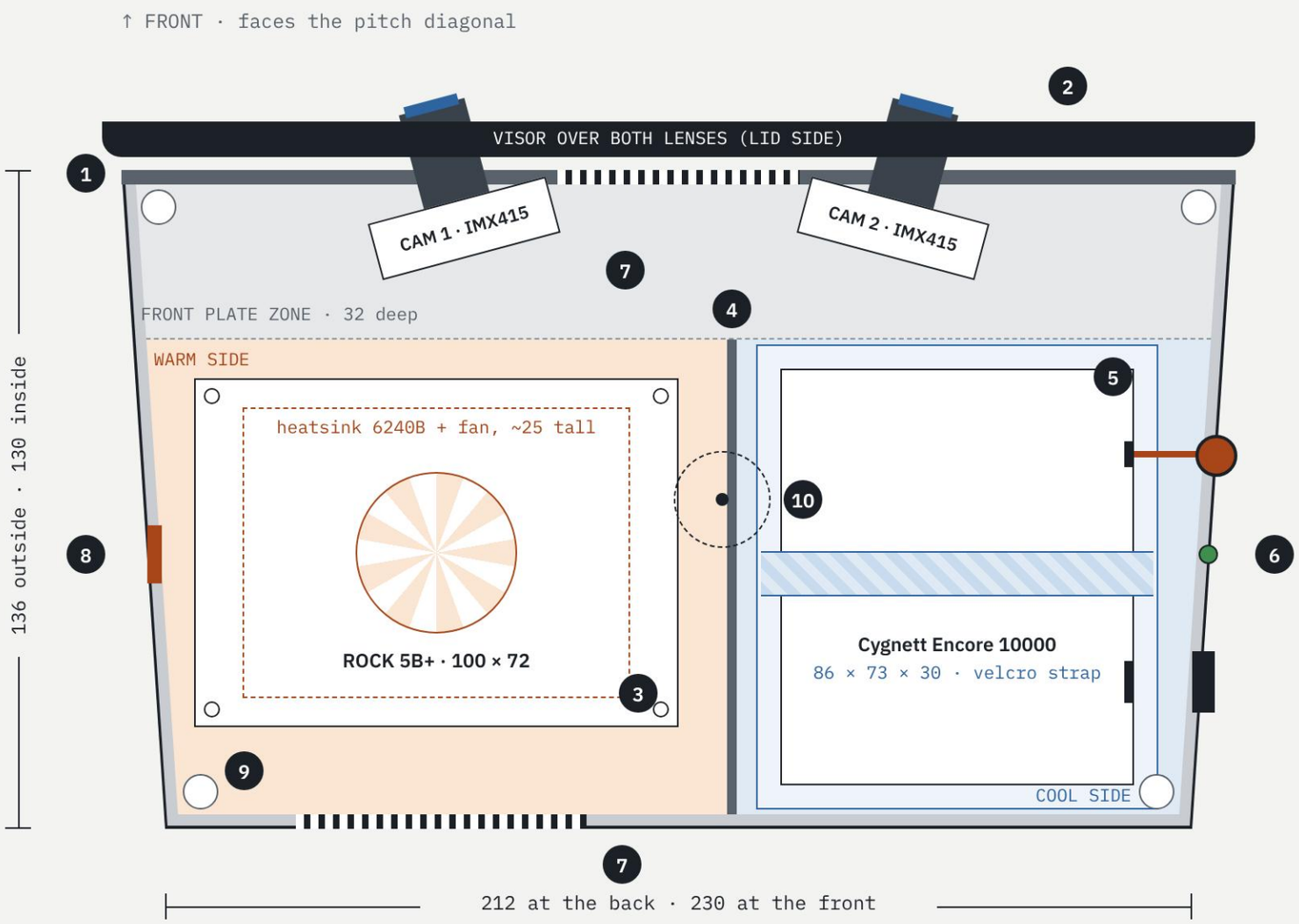
SHEET 01 · PLAN VIEW, LID OFF

Corner camera enclosure

Warm side for the board, cool side for the battery, cameras on a bolt-on front plate, everything you touch on one side wall. The shell tapers 9mm each side towards the back. Drawn at 3 px per mm.



REV	DATE
A1 · yellow wedge	**5 Nov 2026**
INSIDE W (FRONT → BACK)	INSIDE H (FRONT → BACK)
224 → 206 × 130 deep	**62 → 48 at the ridge**
OUTSIDE	MATERIAL · WALLS
230 × 136 × 74 high	**Yellow ASA · 3 / 3 / 4**

Callouts

1. **Front plate** bolts on with 4 × M3 and carries the camera angles: 15° splay and 12° down. Wrong angle means reprinting the plate, not the box.
2. **Two cameras, 30° apart**, 6mm M12 lenses through a black face panel. One visor on the lid covers both.
3. **ROCK 5B+** on 10mm M2.5 standoffs, heatsink and fan on top, 10mm clear all round.
4. **Divider wall** keeps fan heat off the bank. A notch at the top passes the USB-C cable and LED leads.
5. **Battery tray**: loose, foam floor, velcro strap, 5mm clear. Still fits at the narrow back with 4mm to spare.
6. **Touch wall**: push button on a plunger over the bank's button, status LED, covered charge slot.
7. **Vents**: in low on the front plate, out high on the back wall. Both under visors, mesh behind.
8. **microSD access** slot with a cover. Final position once the slot is measured.
9. **Lid**: crowned roof with drip edges, 4 × M3 at the corners, gasket in a printed channel, bubble level on top.
10. **Underside mount**: bracket plate set 3° nose-down, offset towards the battery to sit under the centre of mass.

Warm side · Cool side · Front plate zone · Vent · Heatsink outline

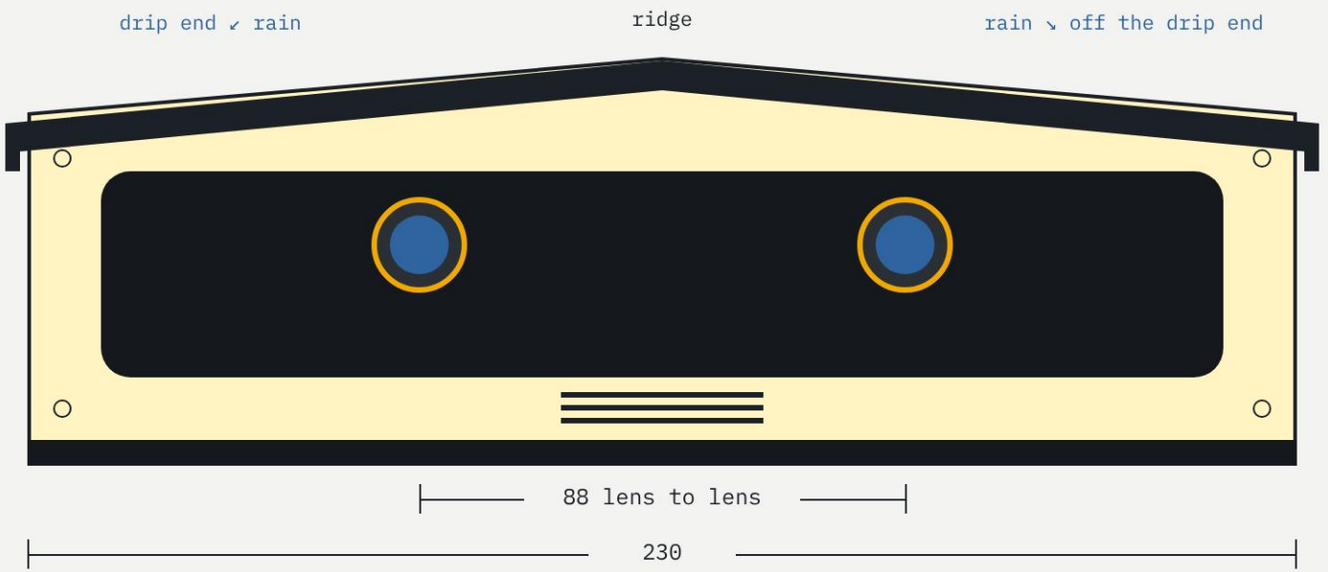
Placeholders until measured: board 100 × 72 · heatsink 25 tall · bank 86 × 73 × 30

SHEET 02 · FRONT PLATE, TILT AND RAIN

Rain runs back and to the sides, never over the lenses

The box sits only 3° nose-down on the bracket. The other 12° of camera tilt is printed into the front plate, so the roof still falls backwards, and its centre ridge sends rain off both sides.

A · Front elevation



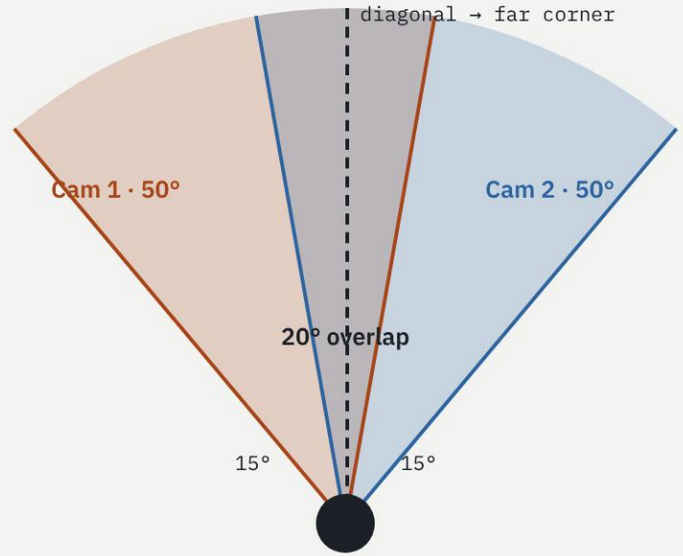
Crowned roof: a centre ridge, 5° fall to each side and 14mm fall from front to back.

Face panel in gloss black. The lenses sit in pads pitched 12° down and splayed 15° (a compound angle).

Visor over both lenses, crowned like the roof. Its raised top edge is a drip dam: water at the front runs along it to the drip ends.

Intake louvres low under the panel, slats angled down, mesh behind.

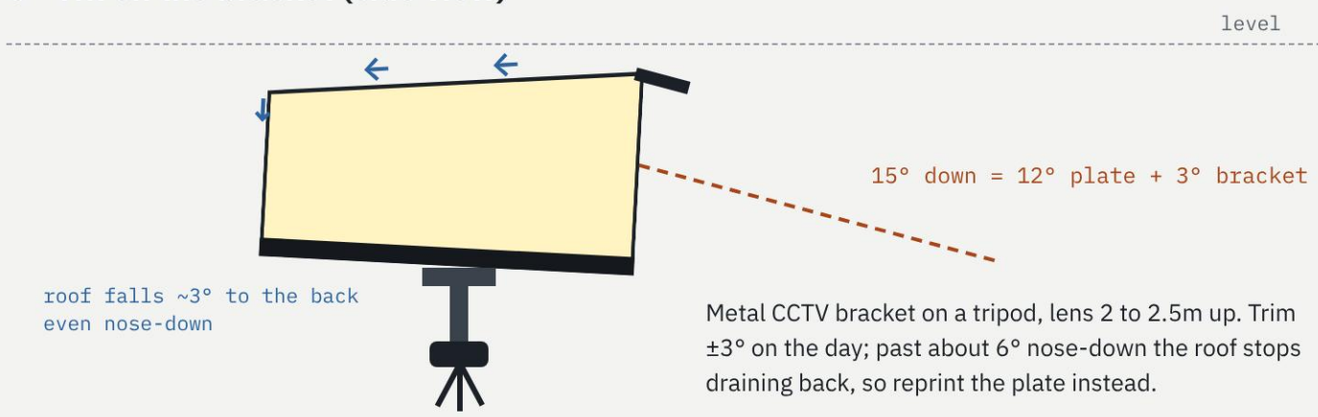
B · Splay, top view



80°	30°	12° + 3°
combined view	splay, in the plate	plate + bracket tilt

Both cameras see the far corner, the distance that sets ball size. A third camera on an 8mm lens for 9v9 and 11v11 would go in a middle slot aimed down the diagonal: leave room in the plate between the pads, but don't drill it yet.

C · Tilt on the bracket (side view)



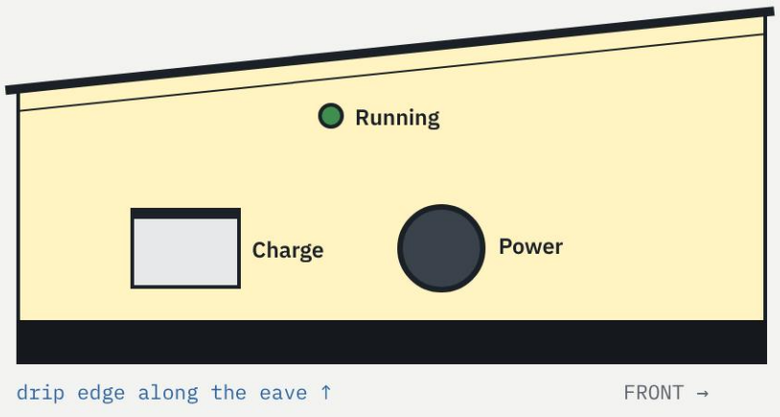
Metal CCTV bracket on a tripod, lens 2 to 2.5m up. Trim ±3° on the day; past about 6° nose-down the roof stops draining back, so reprint the plate instead.

SHEET 03 · TOUCH WALL, AIRFLOW, BATTERY TRAY

Air in low at the front, out high at the back

The roof drops 14mm towards the back and 5° towards each side. There's still room over the fan and the bank, and a drip edge along each eave throws rain clear of the side walls.

A · Touch wall (right side, outside)

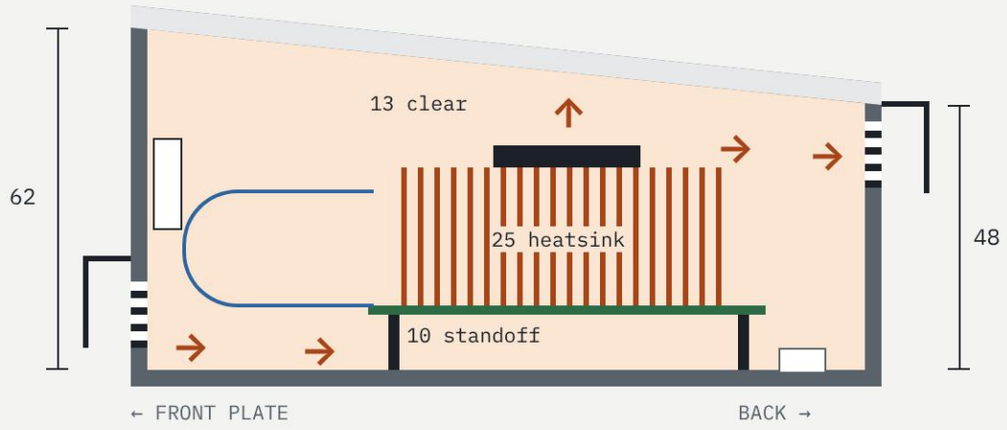


Power: 12 to 16mm waterproof momentary button. Its plunger lands on the bank's own button. Stop recording, wait a few seconds, then press.

Running: 5mm LED on a GPIO pin through 330 Ω, or a clear window over the board's LED.

Charge: hinged cover over the bank's USB-C port. Charge with the lid open, never sealed.

B · Section through the warm side

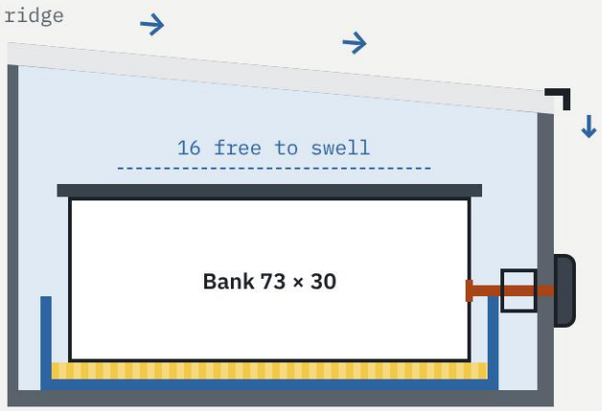


Intake: louvres low in the front plate, slats facing down, mesh behind. **Exhaust**: high on the back wall under a visor.

Sloping lid: 62mm inside at the front, 48mm at the back. The fan still has 13mm above it, more than the 10mm the spec asks for.

Ribbons (blue) loop forward in a gentle curve, clipped clear of the fan. **Silica sachet** on the floor at the back.

C · Battery tray, section from the front



Tray is loose and lifts out by its lip. Foam floor (yellow), velcro strap (dark) over the top through slots in the tray walls.

Plunger runs in a printed guide tube, held in line with the bank's button by the tray walls.

Rain runs off the crown to this side and leaves at the drip edge, clear of the button.

BATTERY RULES

- Never charge it sealed inside the closed box.
- Never charge below 0°C.
- Keep it out of direct sun and away from the heatsink.
- Stop using it if it swells.
- On a hot day, crack the lid between games.

Real measurements are a four-number change

Every size is a variable in the OpenSCAD model. Orange rows are placeholders to replace once the parts are on the bench.

VARIABLE	VALUE	SOURCE
board_w × board_d	100 × 72	placeholder
heatsink_h	25	placeholder
bank_w × d × h	73 × 86 × 30	shop listing
ribbon_len	?	measure AC006
standoff_h	10	spec, 6 to 12
heatsink_clear / bank_clear	10 / 5	spec
cam_splay	±15°	spec
cam_pitch_plate	12°	design
bracket_tilt	3° ± 3°	design
roof_crown	5°	design
roof_fall	14	design
side_taper	9 each	design
visor_reach / drip_dam	20 / 4	design
button_d / led_d	16 / 5	spec
wall / floor / lid	3 / 3 / 4	assumed
lens_spacing	88	derived
outside W × D × H	**230 × 136 × 74**	**derived**

Fixings and bought parts

- 4 × M3 lid screws into heat-set inserts in the corner bosses
- 4 × M3 for the front plate
- M2.5 brass standoffs for the board, M2 for the camera boards
- Gasket cord in the lid channel
- Waterproof momentary button, 12 to 16mm panel mount
- 3 or 5mm LED, panel holder, 330 Ω resistor, two Dupont leads
- Mesh vent cover behind both vents, silica gel sachet
- Velcro strap and thin EVA foam for the tray
- 1/4" tripod insert or a plate for the bracket
- Metal CCTV tilt bracket and a sturdy tripod
- Short USB-C to USB-C, 15 to 30cm, 60W rated

Why 230 wide? The taper and the sloping, crowned roof take space at the back. The box is 12mm wider than the square version, so the board and tray keep their clearances at the narrow back end.

Gates before printing

1. Both cameras at 4K50, H.265, 30Mbps for 60 minutes: files play, card keeps up, chip doesn't throttle.
2. Boots and records 60 minutes on the Cygnett bank, draw measured.
3. Heatsink 6240B confirmed to fit the 5B+, fan spins.
4. 6mm lenses fitted, focused at about 30m, locked.
5. Board, heatsink, bank, ribbons and microSD slot position measured, then the orange rows updated.
6. Hose test on the bracket at 3° nose-down: water leaves by the sides and back, none over the face panel.

Print

- Yellow ASA body, black ASA for the visor, face panel and skirt. ASA handles sun and heat better than PETG.
- Box, lid, front plate, visor and tray print as separate parts.
- After assembly: 60 minutes recording in the garden with the lid on, checking temperatures and the button.

Yellow wedge, crowned roof

23 × 14 × 7.4 cm. The box sits almost level and the cameras do the looking down, so rain leaves by the sides and back, not over the lenses.

Front right | Front | Back left | From above



Roof: a black centre ridge with 5° falls to each side and 14mm down to the back. Black drip edges along the eaves throw water clear of the walls.

Visor: crowned like the roof, over both lenses. Its top edge dams water at the front and sends it to the drip ends.

Tilt: 12° down is built into the lens pads; the bracket adds 3°.

Right side: black power button, green running light, charge flap.

Back: lower than the front, exhaust slots under a small visor.

CONCEPT B · BINOCULAR

Two pods and a rain valley

Two rounded yellow barrels, one per camera, joined by a black bridge. Rain runs off the crowns, gathers in the valley down the middle and leaves by a spout at the back.

Front right | Front | Back left | From above



Left pod: the board and heatsink, camera 1 at the front, intake under its hood, exhaust high on its back.

Right pod: the battery in a curved cradle, camera 2 at the front, button, light and charge flap on its outer side.

Valley: a black channel between the pods, falling to a spout at the back.

Hoods: flared to just clear the lens's 50° × 29° view, splayed 15° and pitched 19° down. The top runs further forward than the bottom, so they double as visors.

Stance: the body sits 4° tail-down so everything drains backwards; the hoods' extra pitch still aims the cameras 15° down.

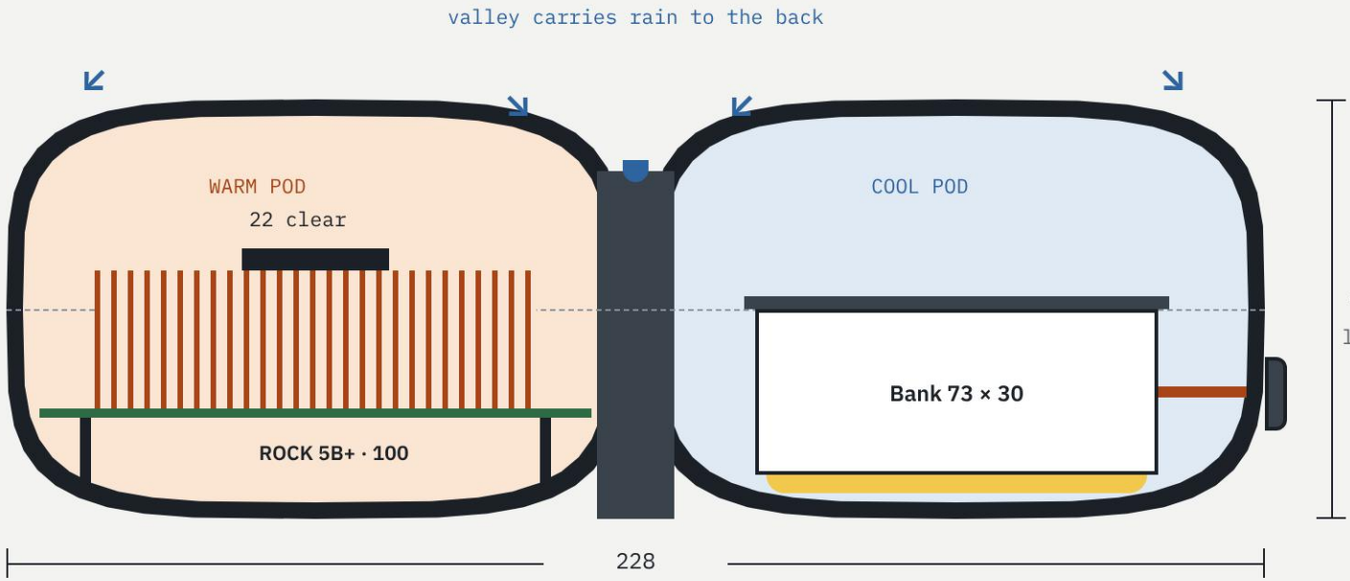
228 × 136 × 78 mm, hoods reach 36 mm ahead · yellow lid, black base · placeholder sizes · shell colour is a tweak

CONCEPT B · SECTIONS AND FIT

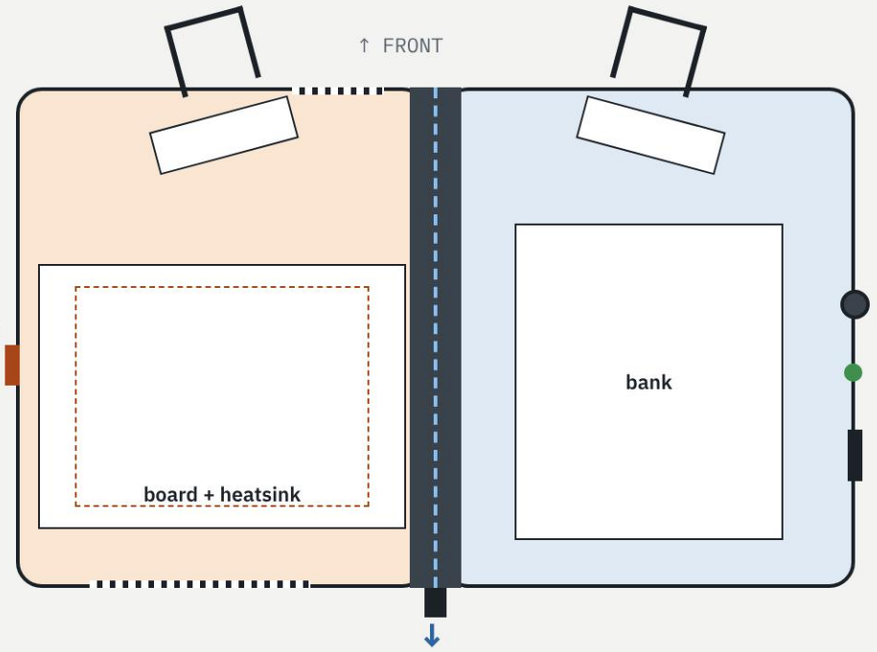
Board in one barrel, battery in the other

Each pod is a rounded-square section, 112 × 76 mm, so a 100mm board still lies flat inside. The lid is the top half of both pods and the valley, split at the middle with one gasket.

A · Section across both pods, from the front

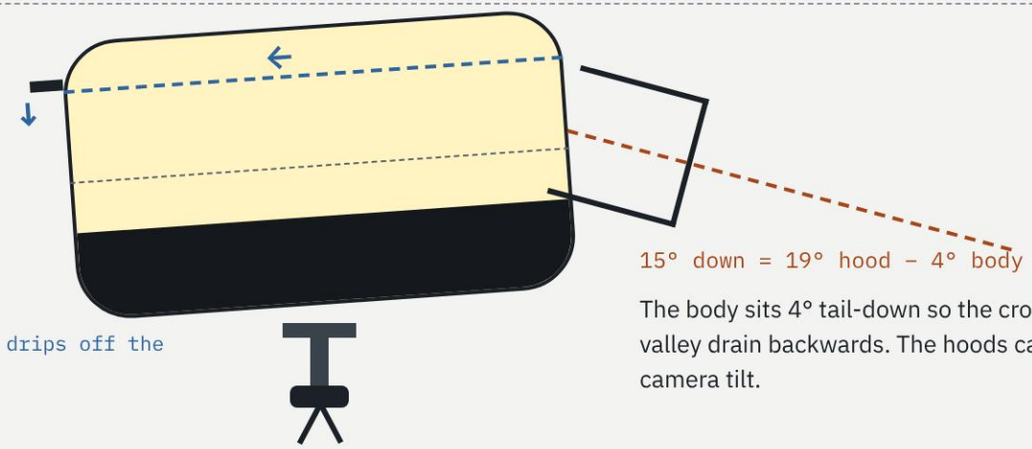


B · Plan, lid off



Lenses 116mm apart. Each hood reaches 36mm ahead of its nose plate, flared to clear the view, with the top longer than the bottom so it works as the visor. Camera 2's ribbon crosses under the valley to the board, about 15cm.

C · Side view on the bracket



The body sits 4° tail-down so the crowns and the valley drain backwards. The hoods carry the camera tilt.

COMPARED WITH THE WEDGE

Better: board and battery sit in separate pods, so the heat stays off the bank without a divider wall. Rain has one clear path, down the valley and out the back.

Power: a printed plunger through a sealed guide presses the bank's own button.

Harder: the 100mm board clears the curved walls by about 2mm at its height, on 12mm standoffs. The battery needs a cradle shaped to the curve. Two curved shells are harder to print than flat walls.

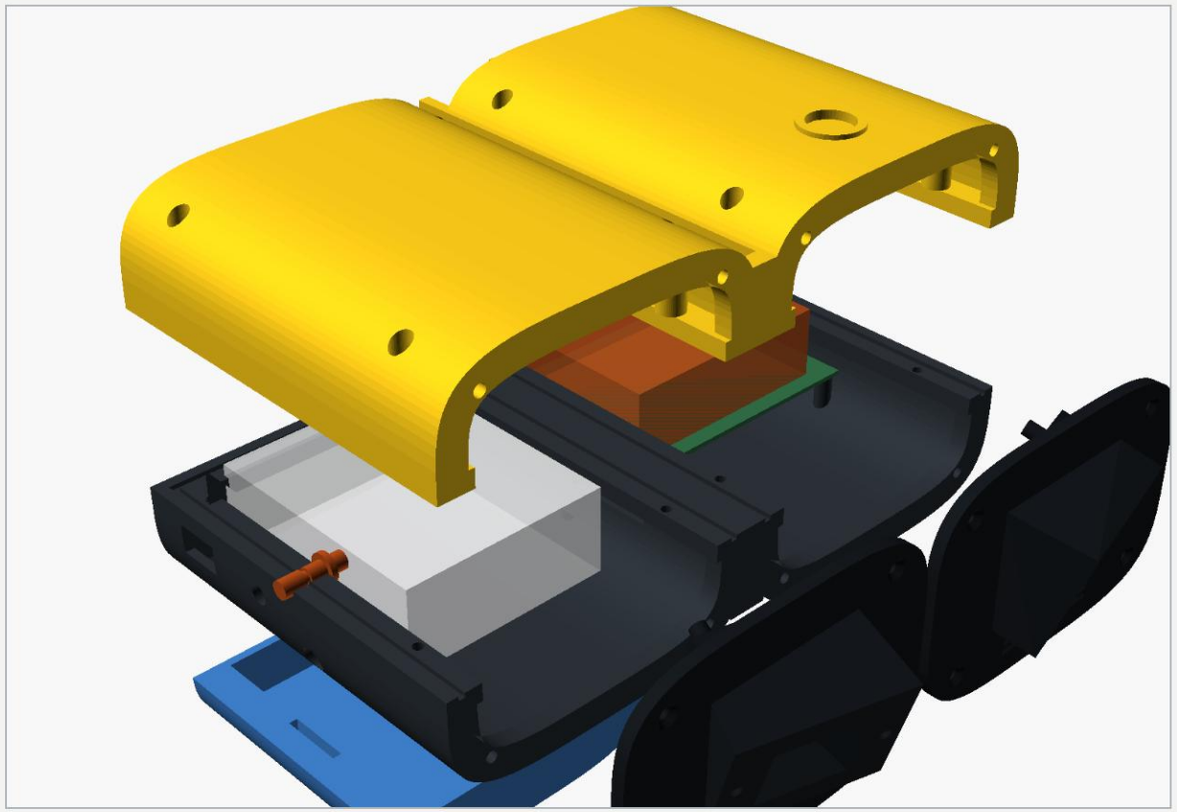
CONCEPT B · OPENSCAD MODEL

The binocular, as modelled

Rendered from the parametric files in `hardware/enclosure-binocular/`. Every part exports as one clean solid at the placeholder sizes.



Assembled: yellow lid, black base, nose plates and hoods



Pulled apart: board and heatsink (left pod), bank on its cradle and the plunger (right pod)

Parts

- **base**: black ASA, board standoffs, plunger guide, tripod pad
- **lid**: yellow ASA, rain channel and spout, exhaust vent, level seat
- **nose_warm / nose_cool**: camera pads at 15° out and 19° down
- **hood** ×2, **cradle**, **small_parts** (plunger, cap, two TPU plugs)

No separate visor

The hoods are the visors. They flare to just clear the 6mm lens's 50° × 29° view, and the top runs 8mm past the bottom so rain drips clear. A groove under the top lip stops water creeping back onto the glass. Remove them for the stock 2.95mm lens.

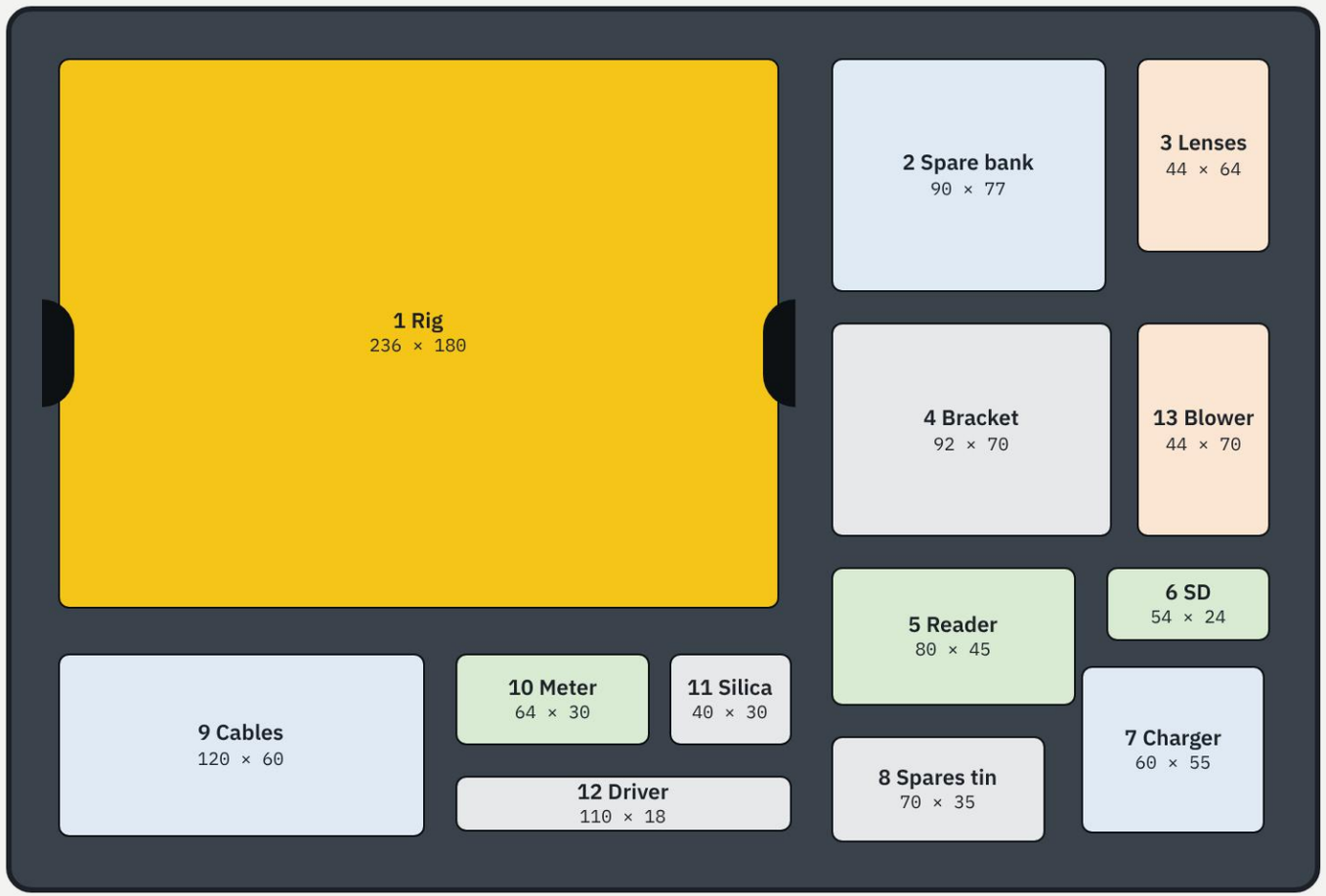
Fit at placeholder sizes

- Board: 2mm clear each side, 26mm above the heatsink
- Cradle walls 4.6mm at the pocket floor
- 228 × 136 × 78mm; hoods reach 36mm ahead
- Measure board holes, bank button and port, camera board and lens position before printing

TRANSPORT · YY430A HARD CASE

Everything for match day in one case, except the tripod

Inside 430 × 290 × 184 mm. Base foam layout, plan view, drawn at 1.7 px per mm. Each pocket is the item plus 2mm all round; depths assume about 140mm of foam in the base.



HINGE SIDE ↑ (top of drawing) 430 × 290 inside HANDLE SIDE ↓

The rig lies on its black base with the hoods facing the hinge, so nothing presses on them when the lid shuts. Black half-moons are finger notches for lifting it out.

Pockets

#	ITEM	POCKET	DEEP
1	Camera rig, hoods on	236 × 180	90 mm
2	Spare Cygnett bank	90 × 77	35 mm
3	M12 lenses, boxed *	44 × 64	45 mm
4	CCTV tilt bracket *	92 × 70	65 mm
5	Icy Box card reader	80 × 45	20 mm
6	Two microSD cards	54 × 24	6 mm
7	PD 30W charger *	60 × 55	35 mm
8	Screws, O-rings, gasket cord	70 × 35	25 mm
9	Cables: USB-C, 12V trigger lead	120 × 60	50 mm
10	USB-C power meter *	64 × 30	30 mm
11	Silica gel sachets	40 × 30	15 mm
12	Screwdriver / hex key	110 × 18	20 mm
13	Lens blower	44 × 70	45 mm

Sizes marked * are guesses: measure the bracket, charger, lens box and power meter before cutting.

Lid foam: lens cloth, printed calibration sheet and pitch marks, clapper. Keep it flat so it presses the rig into its pocket.

Separate bag: tripod (folds longer than the case), 8 cones, 30m tape, line marker.

After rain: dry the rig and leave the case open before shutting it. The case is airtight and traps moisture that fogs lenses.

Cutting: the case's own foam works, but a solid foam sheet cut with a long sharp blade gives cleaner walls than pick-and-pluck.